

assignment_vectors and projectile 1

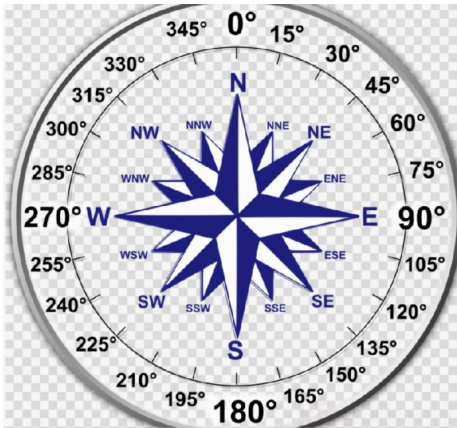
 This is a preview of the published version of the quiz

Started: Apr 29 at 7:29pm

Quiz Instructions

2 attempts/ tutorial available in dropbox. folder HW

See also tutorials about vectors in the pdfs folder in my dropbox



Question 1 10 pts

A velocity vector has components 36 m/s westward and 22 m/s northward. What are the magnitude and direction of this vector?

- "North of west" = starting at west, move toward north (sweeping angle) or NW
- "West of north" = starting at north, move toward west (sweeping angle)



42 m/s @ 30 degrees W of N



58m/s @ 30 degrees NW



42 m/s @ 30 degrees N of W



42 m/s @ 60 degrees N of W



Question 2 10 pts

When rolled down a mountainside at 7.0 m/s, the horizontal component of its velocity vector was 1.8 m/s. What was the angle of the mountain surface above the horizontal?

hint: draw the vector in a polar graph. attach tail to origin. . It is in the 4th quadrant. You are given the magnitude.

☐

75°

☐

33°

☐

15°

☐

57 °

☐

Question 3 20 pts

The x component of vector $\vec{A}$ is 5.3 units, and its y component is -2.3 units. The angle that vector $\vec{A}$ makes with the +x-axis is closest to

☐

23°

☐

250°

☐

110°

☐

160°

☐

340°

☐

Question 4 10 pts

A displacement vector is 34.0 m in length and is directed 60.0° east of north. Selecting from the choices in the table below, what are the components of this vector?

	Northward	Eastward
choice	component	component
1	29.4 m	17.0 m
2	18.2 m	28.1 m
3	22.4 m	11.5 m
4	17.0 m	29.4 m
5	25.2 m	18.2 m

☐

choice 5

☐

choice 3

☐

choice 1

☐

choice 4

☐

choice 2

⋮

Question 5 15 pts

A player throws a football 50.0 m at 61.0° north of west. What is the westward component of the displacement of the football?

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24.2 m

☐

0.00 m

☐

74.0 m

☐

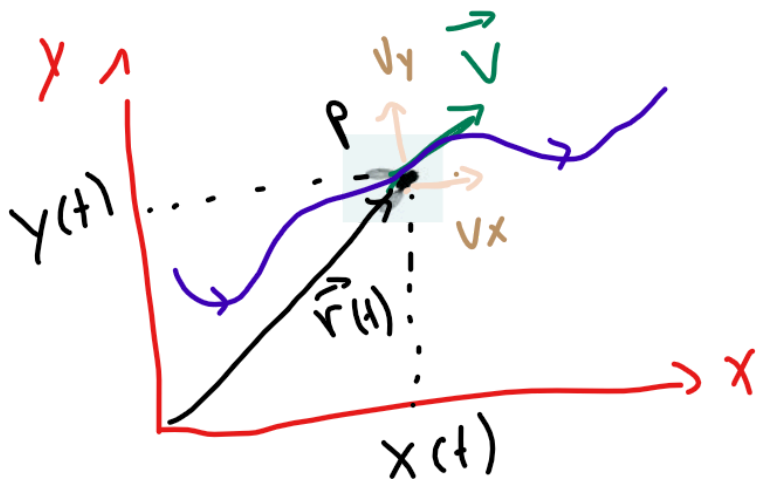
55.0 m

☐

64.7m

⋮

Question 6 15 pts



Imagine an object moving in a 2D plane. Its position is $r(x,y)$. its velocity V has 2 components V_x and V_y .

An object has a position given by $\vec{r} = [2.0 \text{ m} + (5.00 \text{ m/s})t]\hat{i} + [3.0 \text{ m} - (2.00 \text{ m/s}^2)t^2]\hat{j}$, where quantities are in SI units. What is the speed of the object at time $t = 2.00 \text{ s}$?

hints: (hide for first trial)

The object is moving in a plane X-Y. The vector r is the vector position (tail at origin and pointing to the object). This object at a given time has coordinates $r(X(t), Y(t))$ and velocity $V (V_x, V_y)$ and an acceleration (a_x, a_y) .

This means: $X(t)=2+5t$ (position of the object along the axis. the velocity is V_x is constant) and $Y(t)=3-2t^2$ (position of the object along the y-axis . acceleration is -4m/s^2 along Y-axis)

take the derivative of $X(t)$ to get $V_x(t)$ (x-component of the velocity) and the derivative of $Y(t)$ to get $V_y(t)$

The x-component of the velocity at $t=2\text{s}$ is $V_x(2)$ and the y-component of the velocity at $t=2\text{s}$ is $V_y(2)$

TO find the magnitude, use pythagorean theorem. $V = \sqrt{V_x^2 + V_y^2}$

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6.40 m/s

☐

13.0 m/s

☐

9.43 m/s

☐

7.00 m/s

☐

7.65 m/s

⋮

Question 7 10 pts

At the famous Cliff Café, a chef launches a meatball horizontally from the restaurant balcony, which is 20 meters above the ground. The meatball leaves his plate with a speed of 15 m/s. Below, the café's mascot (a hungry dog) waits to catch it.

What is...

(a) The time it takes for the meatball to hit the ground?

(b) The distance from the base of the balcony where the meatball lands (horizontal range)?

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2s, 30m

☐

1.43s, 28.5m

☐

4s, 60m

☐

2s, 15m



Question 8 10 pts

Building from the same question. IF the meat ball is dropped instead from the same height. not launched horizontally.

☐

it will take less time to reach the ground

☐

IDK

☐

it will take the same time to reach the ground

☐

can not be determined



Question 9 10 pts

Chef Luigi launches a meatball horizontally from the balcony of Cliff Café. The meatball has a horizontal velocity of 15 m/s and hits the ground exactly **2.0 seconds** after launch. What far from the cliff the meat ball hits the ground. (horizontal displacement)

☐

10m

☐

20m

☐

30m

☐

40m

Not saved

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